



```
In [78]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (12, 6)

pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", 100)
pd.options.display.float_format = "{:,.2f}".format
```

```
In [79]: file_path = r"C:\Users\Admin\OneDrive\Desktop\My Folder\Data Analytics\Hussair

sheets = pd.read_excel(file_path, sheet_name=None)

print("Available Sheets:")
print(list(sheets.keys()))
```

```
Available Sheets:
['Sales', 'Finance', 'HR', 'Developer Tools']
```

```
In [81]: for sheet_name, df in sheets.items():
    print("\n" + "="*60)
    print(f"SHEET: {sheet_name.upper()}")
    print("="*60)

    print("\nFirst 5 Rows:")
    display(df.head())

    print("\nShape:")
    print(df.shape)

    print("\nData Types:")
    print(df.dtypes)

    print("\nMissing Values:")
    print(df.isnull().sum())

    print("\nDuplicate Rows:")
    print(df.duplicated().sum())
```

```
=====
SHEET: SALES
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```

```
First 5 Rows:
```

	Product Name	Category	Business Function	Industry	Pricing Model	Deployment Type	Primary Market
0	Zoho CRM	Sales	Customer Management	Enterprise	Premium	Hybrid	Asia
1	Zoho SalesIQ	Sales	Automation	Retail	Premium	Hybrid	Africa
2	Zoho CRM	Sales	Customer Management	SME	Premium	Hybrid	Asia
3	Zoho Bookings	Sales	Communication	Enterprise	Premium	Cloud	Middle East
4	Zoho Campaigns	Sales	Analytics	Healthcare	Premium	Cloud	Middle East

Shape:
(3500, 14)

Data Types:

Product Name	str
Category	str
Business Function	str
Industry	str
Pricing Model	str
Deployment Type	str
Primary Market	str
Country	str
Integration Level	str
LaunchYear	int64
Active Users Millions	float64
Customer Rating	float64
Support Level	str
Year	int64

dtype: object

Missing Values:

Product Name	0
Category	0
Business Function	0
Industry	0
Pricing Model	0
Deployment Type	0
Primary Market	0
Country	0
Integration Level	0
LaunchYear	0
Active Users Millions	0
Customer Rating	0
Support Level	0
Year	0

dtype: int64

Duplicate Rows:
0

=====

SHEET: FINANCE

=====

First 5 Rows:

	Product Name	Category	Business Function	Industry	Pricing Model	Deployment Type	Primary Market
0	Zoho Billing	Finance	Communication	Manufacturing	Premium	Cloud	Middle East
1	Zoho Books	Finance	Accounting	Finance	Premium	Cloud	Europe
2	Zoho Billing	Finance	Accounting	Healthcare	Premium	Cloud	Asia
3	Zoho Books	Finance	Automation	Healthcare	Free	Cloud	North America
4	Zoho Invoice	Finance	Automation	SME	Subscription	Hybrid	North America

Shape:
(3500, 14)

Data Types:

Product Name	str
Category	str
Business Function	str
Industry	str
Pricing Model	str
Deployment Type	str
Primary Market	str
Country	str
Integration Level	str
LaunchYear	int64
Active Users Millions	float64
Customer Rating	float64
Support Level	str
Year	int64

dtype: object

Missing Values:

Product Name	0
Category	0
Business Function	0
Industry	0
Pricing Model	0
Deployment Type	0
Primary Market	0
Country	0
Integration Level	0
LaunchYear	0
Active Users Millions	0
Customer Rating	0
Support Level	0
Year	0

dtype: int64

Duplicate Rows:
0

=====

SHEET: HR

=====

First 5 Rows:

	Product Name	Category	Business Function	Industry	Pricing Model	Deployment Type	Primary Market
0	Zoho Workerly	HR	Security	Finance	Premium	Cloud	Middle East
1	Zoho People	HR	Automation	Finance	Free	Cloud	Asia
2	Zoho Shifts	HR	Customer Management	SME	Pay-as-you-go	Cloud	Middle East
3	Zoho People	HR	Automation	Manufacturing	Free	Cloud	Asia
4	Zoho Recruit	HR	Automation	Healthcare	Pay-as-you-go	Cloud	Europe

Shape:
(3500, 14)

Data Types:
Product Name str
Category str
Business Function str
Industry str
Pricing Model str
Deployment Type str
Primary Market str
Country str
Integration Level str
LaunchYear int64
Active Users Millions float64
Customer Rating float64
Support Level str
Year int64
dtype: object

Missing Values:
Product Name 0
Category 0
Business Function 0
Industry 0
Pricing Model 0
Deployment Type 0
Primary Market 0
Country 0
Integration Level 0
LaunchYear 0
Active Users Millions 0
Customer Rating 0
Support Level 0
Year 0
dtype: int64

Duplicate Rows:
0

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SHEET: DEVELOPER TOOLS

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First 5 Rows:

	Product Name	Category	Business Function	Industry	Pricing Model	Deployment Type	Primary Market
0	Zoho API	Developer Tools	Customer Management	Healthcare	Subscription	Hybrid	Europe
1	Zoho Catalyst	Developer Tools	Automation	Enterprise	Subscription	Cloud	Asia
2	Zoho Catalyst	Developer Tools	Accounting	Finance	Freemium	Hybrid	Europe
3	Zoho Flow	Developer Tools	Security	Education	Subscription	Hybrid	South America
4	Zoho DataPrep	Developer Tools	Customer Management	SME	Freemium	Hybrid	Middle East

Shape:
(3500, 14)

Data Types:

Product Name	str
Category	str
Business Function	str
Industry	str
Pricing Model	str
Deployment Type	str
Primary Market	str
Country	str
Integration Level	str
LaunchYear	int64
Active Users Millions	float64
Customer Rating	float64
Support Level	str
Year	int64

dtype: object

Missing Values:

Product Name	0
Category	0
Business Function	0
Industry	0
Pricing Model	0
Deployment Type	0
Primary Market	0
Country	0
Integration Level	0
LaunchYear	0
Active Users Millions	0
Customer Rating	0
Support Level	0
Year	0

dtype: int64

Duplicate Rows:
0

```
In [82]: for sheet_name, df in cleaned_sheets.items():
          print("\n" + "="*60)
          print(f"CLEANED SHEET: {sheet_name.upper()}")
          print("="*60)

          print(df.info())
          print("\nSummary Statistics:")
          print(df.describe())
```

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CLEANED SHEET: SALES

=====

```
<class 'pandas.DataFrame'>
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   product_name                          3500 non-null   str
1   category                              3500 non-null   str
2   business_function                     3500 non-null   str
3   industry                              3500 non-null   str
4   pricing_model                         3500 non-null   str
5   deployment_type                       3500 non-null   str
6   primary_market                        3500 non-null   str
7   country                               3500 non-null   str
8   integration_level                     3500 non-null   str
9   launchyear                            3500 non-null   int64
10  active_users_millions                  3500 non-null   float64
11  customer_rating                       3500 non-null   float64
12  support_level                          3500 non-null   str
13  year                                   3500 non-null   int64
dtypes: float64(2), int64(2), str(10)
memory usage: 659.4 KB
None
```

Summary Statistics:

	launchyear	active_users_millions	customer_rating	year
count	3,500.00	3,500.00	3,500.00	3,500.00
mean	2,014.57	75.03	4.25	2,024.01
std	5.74	43.19	0.44	0.82
min	2,005.00	0.11	3.50	2,023.00
25%	2,010.00	37.56	3.90	2,023.00
50%	2,015.00	75.97	4.30	2,024.00
75%	2,019.00	113.11	4.60	2,025.00
max	2,024.00	150.00	5.00	2,025.00

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CLEANED SHEET: FINANCE

=====

```
<class 'pandas.DataFrame'>
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   product_name                          3500 non-null   str
1   category                              3500 non-null   str
2   business_function                     3500 non-null   str
3   industry                              3500 non-null   str
4   pricing_model                         3500 non-null   str
5   deployment_type                       3500 non-null   str
6   primary_market                        3500 non-null   str
7   country                               3500 non-null   str
8   integration_level                     3500 non-null   str
```

9	launchyear	3500 non-null	int64
10	active_users_millions	3500 non-null	float64
11	customer_rating	3500 non-null	float64
12	support_level	3500 non-null	str
13	year	3500 non-null	int64

dtypes: float64(2), int64(2), str(10)

memory usage: 667.2 KB

None

Summary Statistics:

	launchyear	active_users_millions	customer_rating	year
count	3,500.00	3,500.00	3,500.00	3,500.00
mean	2,014.54	74.56	4.25	2,024.02
std	5.76	43.72	0.44	0.81
min	2,005.00	0.15	3.50	2,023.00
25%	2,009.00	35.73	3.90	2,023.00
50%	2,015.00	74.89	4.30	2,024.00
75%	2,020.00	111.94	4.60	2,025.00
max	2,024.00	149.88	5.00	2,025.00

CLEANED SHEET: HR

<class 'pandas.DataFrame'>

RangeIndex: 3500 entries, 0 to 3499

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	product_name	3500 non-null	str
1	category	3500 non-null	str
2	business_function	3500 non-null	str
3	industry	3500 non-null	str
4	pricing_model	3500 non-null	str
5	deployment_type	3500 non-null	str
6	primary_market	3500 non-null	str
7	country	3500 non-null	str
8	integration_level	3500 non-null	str
9	launchyear	3500 non-null	int64
10	active_users_millions	3500 non-null	float64
11	customer_rating	3500 non-null	float64
12	support_level	3500 non-null	str
13	year	3500 non-null	int64

dtypes: float64(2), int64(2), str(10)

memory usage: 648.9 KB

None

Summary Statistics:

	launchyear	active_users_millions	customer_rating	year
count	3,500.00	3,500.00	3,500.00	3,500.00
mean	2,014.50	74.74	4.24	2,023.99
std	5.74	43.13	0.43	0.82
min	2,005.00	0.17	3.50	2,023.00
25%	2,010.00	37.61	3.90	2,023.00
50%	2,014.00	74.55	4.20	2,024.00

75%	2,019.00	111.48	4.60	2,025.00
max	2,024.00	149.98	5.00	2,025.00

```
=====
CLEANED SHEET: DEVELOPER TOOLS
=====
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   product_name                          3500 non-null   str
1   category                              3500 non-null   str
2   business_function                     3500 non-null   str
3   industry                              3500 non-null   str
4   pricing_model                         3500 non-null   str
5   deployment_type                       3500 non-null   str
6   primary_market                        3500 non-null   str
7   country                               3500 non-null   str
8   integration_level                     3500 non-null   str
9   launchyear                            3500 non-null   int64
10  active_users_millions                 3500 non-null   float64
11  customer_rating                       3500 non-null   float64
12  support_level                         3500 non-null   str
13  year                                  3500 non-null   int64
dtypes: float64(2), int64(2), str(10)
memory usage: 689.8 KB
None
```

```
Summary Statistics:
      launchyear  active_users_millions  customer_rating  year
count    3,500.00             3,500.00           3,500.00  3,500.00
mean      2,014.40              74.86             4.25  2,023.99
std         5.75              42.98             0.43    0.82
min      2,005.00               0.12             3.50  2,023.00
25%      2,009.00              37.68             3.90  2,023.00
50%      2,014.00              74.61             4.20  2,024.00
75%      2,019.00             111.86             4.60  2,025.00
max      2,024.00             149.97             5.00  2,025.00
```

```
In [83]: sales_df = cleaned_sheets["Sales"]

print("\nColumns in Sales Sheet:")
print(sales_df.columns)
```

```
Columns in Sales Sheet:
Index(['product_name', 'category', 'business_function', 'industry',
      'pricing_model', 'deployment_type', 'primary_market', 'country',
      'integration_level', 'launchyear', 'active_users_millions',
      'customer_rating', 'support_level', 'year'],
      dtype='str')
```

```
In [84]: top_products = (
    sales_df.groupby("product_name")["active_users_millions"]
```

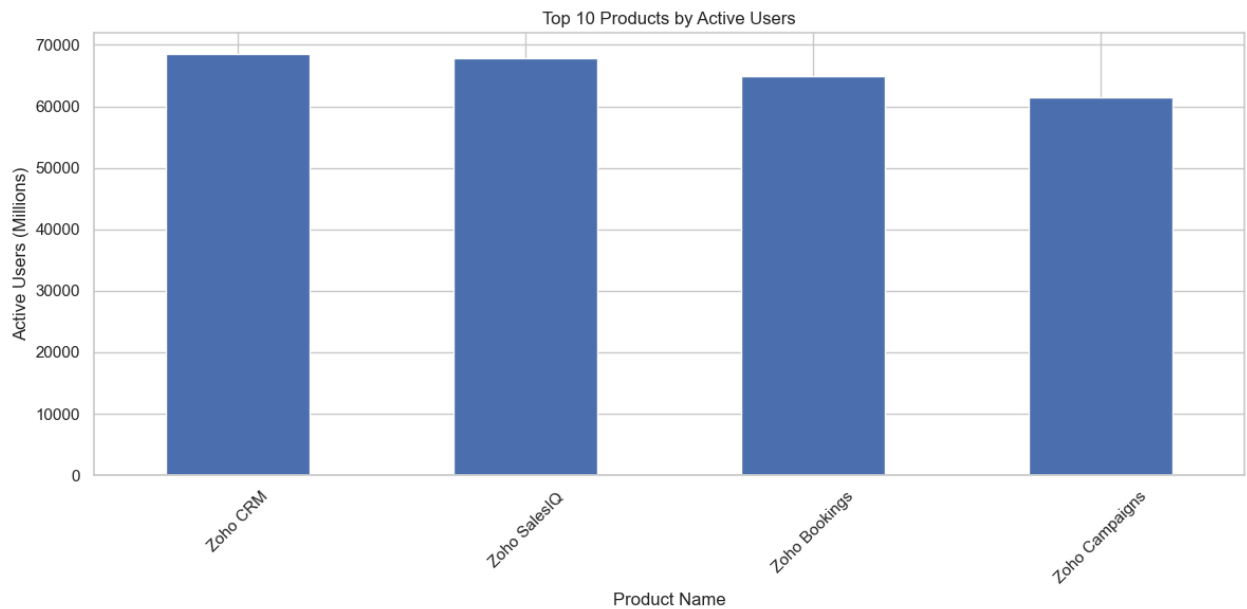
```

        .sum()
        .sort_values(ascending=False)
        .head(10)
    )

    top_products.plot(kind="bar")

    plt.title("Top 10 Products by Active Users")
    plt.xlabel("Product Name")
    plt.ylabel("Active Users (Millions)")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

```



```

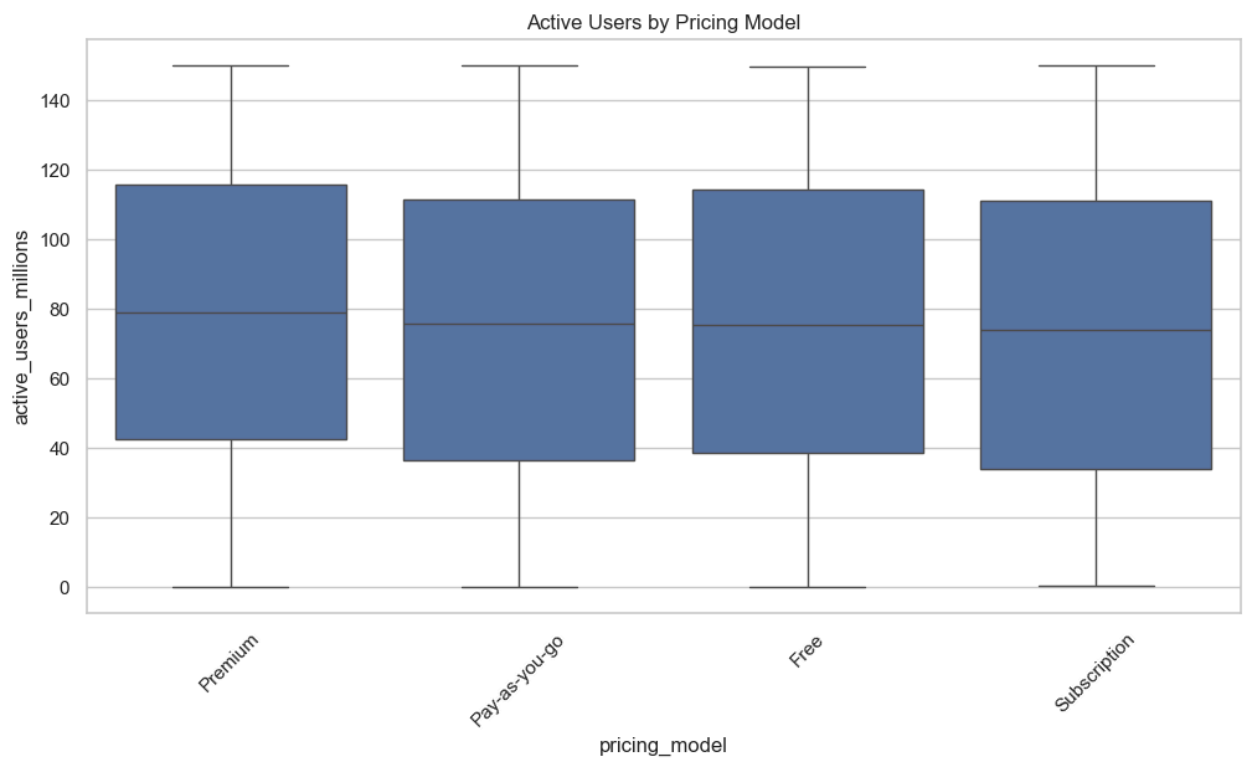
In [68]: sns.histplot(
    sales_df["customer_rating"],
    bins=10,
    kde=True
)

plt.title("Customer Rating Distribution")
plt.xlabel("Customer Rating")
plt.ylabel("Frequency")
plt.show()

```



```
In [69]: sns.boxplot(  
    x="pricing_model",  
    y="active_users_millions",  
    data=sales_df  
)  
  
plt.title("Active Users by Pricing Model")  
plt.xticks(rotation=45)  
plt.show()
```



```
In [70]: all_data = pd.concat(
            cleaned_sheets.values(),
            ignore_index=True
        )

        print(all_data.head())
```

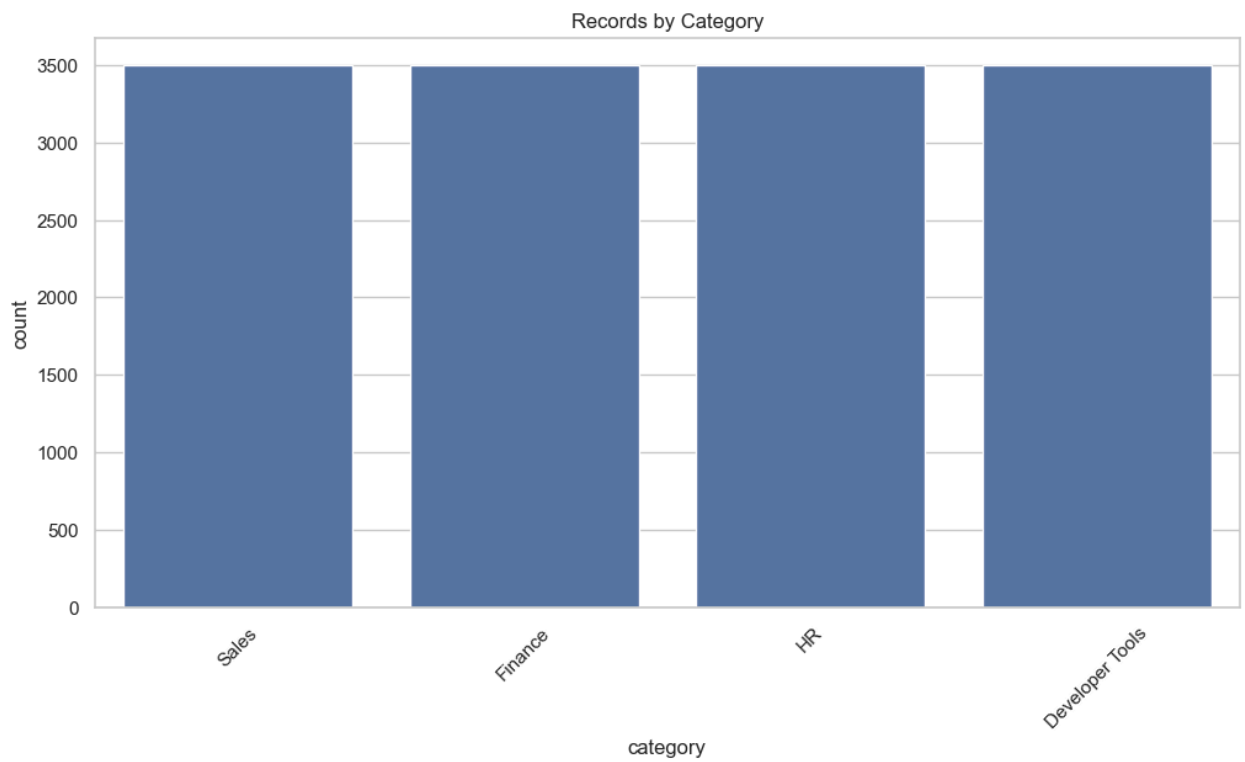
	product_name	category	business_function	industry	pricing_model	\
0	Zoho CRM	Sales	Customer Management	Enterprise	Premium	
1	Zoho SalesIQ	Sales	Automation	Retail	Premium	
2	Zoho CRM	Sales	Customer Management	SME	Premium	
3	Zoho Bookings	Sales	Communication	Enterprise	Premium	
4	Zoho Campaigns	Sales	Analytics	Healthcare	Premium	

	deployment_type	primary_market	country	integration_level	launchyear	\
0	Hybrid	Asia	Russia	High	2022	
1	Hybrid	Africa	Vietnam	High	2019	
2	Hybrid	Asia	Afghanistan	Low	2016	
3	Cloud	Middle East	Uzbekistan	High	2010	
4	Cloud	Middle East	Japan	High	2007	

	active_users_millions	customer_rating	support_level	year
0	136.56	3.80	Standard	2023
1	27.71	4.20	Standard	2024
2	46.77	3.70	Premium	2024
3	43.84	3.70	Enterprise	2025
4	123.30	4.10	Premium	2025

```
In [76]: sns.countplot(
            x="category",
            data=all_data
        )

        plt.title("Records by Category")
        plt.xticks(rotation=45)
        plt.show()
```



```
In [77]: numeric_cols = all_data.select_dtypes(include=np.number)

sns.heatmap(
    numeric_cols.corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")
plt.show()
```

